

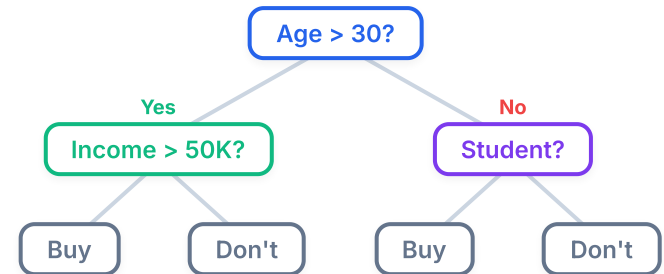
Decision Trees

How machines learn to make decisions using simple yes/no questions.

Day 13 of 100 Days of Machine Learning

What is a Decision Tree?

A **Decision Tree** is a supervised machine learning algorithm that predicts outcomes by repeatedly asking the best possible questions.



How It Works



Splitting Criteria

Gini Index

- Faster
- Computationally efficient
- Used by CART

$$1 - \sum(p^2)$$

Entropy

- Info theory
- Measures uncertainty
- Used by ID3

$$-\sum(p \log_2 p)$$

Rule of thumb: Gini → Faster | Entropy → More informative

Tree Depth



Underfitting

Shallow Tree



Best Generalization

Balanced Tree



Overfitting

Very Deep Tree

Pros & Cons

Advantages ✓

- ✓ Easy to understand
- ✓ No scaling required
- ✓ Nonlinear relationships
- ✓ Classification & regression
- ✓ Feature importance

Disadvantages ✗

- ✗ Can easily overfit
- ✗ Sensitive to noise
- ✗ Unstable to small changes
- ✗ Greedy algorithm

Top Interview Questions

- What is a Decision Tree?
- Difference between Gini and Entropy?
- Why do Decision Trees overfit?
- What is pruning?
- What is max_depth?



Pro Tip: A Decision Tree asks questions that maximize information gain, making each split as "pure" as possible.